PROJECT SYNOPSIS & TECHNICAL SPECIFICATION

A DISSERTATION REPORT SUBMITTED IN PARTIAL FULFILLMENT OF DEGREE REQUIREMENTS

HALL KHOJ FINDER

A MOBILE-FIRST MULTI-VENDOR EVENT VENUE & SERVICES MARKETPLACE WITH 3D SPATIAL VIRTUAL TOURS

ACADEMIC CURRICULUM METADATA

Course / Degree:	**Bachelor of Technology (B.Tech) / BCA / MCA**
Department:	**Department of Computer Science & Engineering**
Domain Specialization:	**Full-Stack Web Development, 3D Computer Graphics & Cloud DB**
Academic Session:	**2025 - 2026**
Live Demonstration URL:	**https://hallkhoj.com • Port: 3000**
Database & Storage:	**MongoDB Atlas + Atomic JSON Failover Storage**

SUBMITTED BY:

Project Development Team

Hall Khoj Project Group

Reg No: [University Roll No]

Final Year Major Project

Candidate Signature: _____

UNDER THE GUIDANCE OF:

Project Guide & Faculty

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2. INTRODUCTION

In India, organizing weddings, recepti
investment. The Indian wedding and e
immense size, the sector remains
recommendations.

Hall Khoj Finder is an enterprise-gra
services ecosystem across India. Th
venues, inspect hall dimensions via
bookings through transparent paymen

3. PROJECT OVERVIEW

Hall Khoj Finder operates as a multi-sided digital marketplace connecting three distinct stakeholders:

- **1. Event Seekers (Customers):** Individuals and families searching for marriage halls, banquet halls, open lawns, catering, and event vendors with complete price and spatial transparency.
- **2. Venue Owners & Event Vendors:** Business operators seeking digital visibility, automated calendar management to avoid double-bookings, and verified partner credibility.
- **3. Platform Administrators (Superadmin):** Regulators overseeing platform health, vendor KYC verifications, financial settlements, and content moderation.

The application is built on a Mobile-First architectural philosophy, specifically tailored for the Indian demographic where over 85% of internet traffic originates from smartphones. It eliminates common digital onboarding friction by utilizing mobile phone numbers (+91) as primary identifiers instead of mandatory email addresses for customers, while enforcing strict email and document verification for commercial venue owners.

4. PROBLEM STATEMENT

The traditional venue booking and event planning ecosystem in India suffers from severe structural bottlenecks:

- **Problem 1 (Opaque Pricing):** Customers routinely pay 10% to 20% inflated pricing because local unverified agents and brokers gatekeep contact details.
- **Problem 2 (High Digital Friction):** Mainstream western event portals enforce strict email-based signups. In Indian regional markets, forcing email verification leads to over 65% drop-off during onboarding.
- **Problem 3 (Misleading Imagery):** Two-dimensional, heavily filtered photos uploaded to social media do not reveal pillar obstructions, ceiling heights, stage elevations, or dining partitions, causing disappointment on the event day.
- **Problem 4 (Catastrophic Double-Booking):** Venue owners relying on traditional paper diaries and verbal promises frequently double-book venues during peak auspicious wedding dates (Muhurat).
- **Problem 5 (Fragmented Allied Services):** Finding a venue is only one part of the event; hiring reliable caterers, tent houses, photographers, and decorators requires separate, uncoordinated negotiation.

5. PURPOSE OF THE PROJECT

The primary purpose of Hall Khoj Finder is to build a trusted, transparent, and technology-driven marketplace that:

- **Direct Connection:** Eliminates unauthorized middlemen and connects consumers directly with venue owners.
- **Spatial Clarity:** Delivers complete spatial and visual clarity through browser-based 3D virtual walkthroughs before physical site visits.
- **Zero Double-Booking:** Guarantees zero double-booking through synchronized real-time availability calendars.
- **Zero-Friction Access:** Minimizes user onboarding friction by replacing email dependencies with high-trust mobile phone authentication for consumers.
- **Vendor Storefronts:** Equips venue owners and micro-entrepreneurs across 11 service categories with professional digital storefronts.

6. OBJECTIVES

- **Zero-Friction Authentication:** Implement a mobile phone-first authentication system for customers and a KYC-verified onboarding pipeline for venue owners.
- **Interactive 3D Virtual Tour:** Implement GPU-accelerated 3D WebGL rendering using Three.js to provide panoramic spatial visualization of banquet layouts.
- **Geospatial Proximity Discovery:** Integrate interactive mapping (Leaflet.js / OpenStreetMap) to allow users to discover venues within defined distance radii.
- **Automated Availability & Booking Engine:** Provide real-time date availability checking with one-click date blockers for venue owners to prevent schedule clashes.
- **Multi-Category Breadth:** Support 11 distinct event domains ranging from Marriage Halls and Resorts to Catering, Photography, and Decorators.
- **Dynamic Invoicing Pipeline:** Automatically generate official PDF invoices and booking receipts on the server using vector graphics engines.
- **High-Performance Architecture:** Maintain sub-600ms load times and high Lighthouse performance on mobile devices over 3G/4G networks.

7. SCOPE OF THE PROJECT

A. In-Scope (Implemented & Fully Operational)

- **Responsive Web App:** Mobile-first PWA-ready responsive web application.
- **Dual Auth Engine:** Mobile phone for customers; Business email & password for venue owners.
- **11 Categories:** Full-spectrum catalog support across 11 verified business categories.
- **3D Walkthrough:** Interactive 3D WebGL venue walkthrough viewer powered by Three.js.
- **GPS Discovery:** Leaflet GPS venue discovery with geolocation radius filtering.
- **Real-Time Calendar:** Date availability checker and owner offline date-blocker tool.
- **Payment Checkout:** Advance token calculation and Razorpay payment gateway integration.
- **Server-Side PDF:** Automated server-side PDF booking receipt generation using PDFKit.
- **Dual Database:** MongoDB Atlas Cloud + Local Resilient JSON cache failover.

B. Out-of-Scope (Future Enhancements)

- **Native App Store Packages:** Dedicated iOS and Android binaries (currently operates via mobile web).
- **Multi-Stage Escrow Automated Payouts:** Instant bank virtual account disbursements (currently advance token verification).

8. EXISTING SYSTEM VS. PROPOSED SYSTEM

PARAMETER	EXISTING TRADITIONAL SYSTEM	PROPOSED (HALL KHOJ FINDER)
Primary Identifier	Email address mandatory (Causes high drop-off)	Mobile Phone (+91) with instant login
Pricing Transparency	Hidden; negotiable via middlemen brokers	Clear per-plate (Veg/Non-Veg) & rental rates
Venue Visualization	Static, misleading, heavily edited 2D photos	Interactive 3D WebGL walkthrough + High-res gallery
Availability Check	Manual phone calls or physical diary check	Live real-time green/red calendar with 1-click blocker
Middleman Commission	10% - 20% unorganized broker cuts	Direct customer-to-vendor connection (0% broker cut)
Vendor Credibility	Unverified word-of-mouth claims	Shield Verified Partner badge with GST/PAN KYC
Receipt / Invoicing	Handwritten paper slips or verbal agreements	Instant server-generated PDF booking invoice
Device Optimization	Heavy desktop-oriented web frameworks	Lightweight Vanilla JS/CSS (<180 KB page weight)

9. TECHNOLOGIES USED & ARCHITECTURE STACK

TECHNOLOGY LAYER	LIBRARY / VERSION	PURPOSE IN PROJECT
Programming Language	JavaScript (ES6+ / Node.js v24.x)	Full-stack language for frontend and backend logic.
Frontend Core	HTML5, Vanilla CSS3, Modern ES6+	Semantic structure, custom design system, fast DOM.
3D Graphics Engine	Three.js (WebGL r128 CDN)	GPU-accelerated 3D panoramic virtual hall tour.
Mapping Engine	Leaflet.js (v1.9.4 CDN)	Vector map pins, proximity radius, OpenStreetMap.
Backend Framework	Express.js (^5.2.1)	Asynchronous RESTful API server and routing engine.
Database (Primary)	MongoDB & Mongoose (^9.10.2)	Document database for entity schemas and geo-queries.
Database (Fallback)	Atomic JSON File Store (Native fs)	Local storage engine for zero-downtime failover.
Authentication	JSON Web Token (JWT ^9.0.3)	Stateless HS256-signed Bearer tokens for API auth.
Password Encryption	Bcryptjs (^3.0.3)	10-round salted Blowfish cryptographic password hashing.
File Upload Engine	Multer (^2.4.0)	Multipart form-data parser for cover and gallery photos.
PDF Generator	PDFKit (^0.20.2)	Programmatic vector PDF generator for receipts & synopsis.
Payment Gateway	Razorpay / Cashfree REST SDK	UPI, Net Banking, and Card checkout with HMAC-SHA256.
AI Matchmaker	Google Gemini AI REST API	Natural language event query parsing & recommendations.

10. SYSTEM REQUIREMENTS

A. Hardware Requirements

COMPONENT	MINIMUM CLIENT REQUIREMENTS	RECOMMENDED SERVER SPECIFICATIONS
Processor	Dual-Core 1.5 GHz (Mobile or PC)	Quad-Core 2.4 GHz (x86_64 / ARM64 Cloud CPU)
System RAM	2 GB RAM (Capable of WebGL execution)	4 GB to 8 GB DDR4 RAM
Hard Disk Space	100 MB free space (Browser cache)	20 GB SSD Cloud Storage
Display Resolution	360 x 640 px (Smartphone viewports)	1920 x 1080 px (Superadmin Workstations)
Network Bandwidth	3G / 4G / Wi-Fi (Minimum 512 Kbps)	High-speed Cloud connection (1 Gbps uplink)

B. Software Requirements

SOFTWARE COMPONENT	SPECIFICATION / REQUIREMENT
Operating System (Development)	Windows 10/11, macOS Sequoia, or Linux Ubuntu 22.04 LTS
Backend Runtime Environment	Node.js (v18.x to v24.x LTS) with npm package manager
Database Server	MongoDB Community Server (v6.0+) or MongoDB Atlas Cloud Cluster
Supported Web Browsers	Google Chrome (v90+), Mozilla Firefox, Apple Safari, Microsoft Edge
API Architecture & Standards	RESTful JSON over HTTP/1.1 and secure HTTPS (TLS 1.3 encryption)

C. Architecture Justification for Viva

Viva Note: Unlike standard academic projects that rely on heavy frontend frameworks like Rea intentionally uses pure Vanilla JavaScript and custom CSS custom properties (variables) paired wit the base page payload to under 180 KB, enabling sub-600ms load times on Indian mobile 3G/4G ne application (SPA) bundles cause severe latency. Furthermore, the backend implements an a mechanism between MongoDB Atlas and a local atomic JSON store to prevent system halting during

11. MODULES OF THE PROJECT

Module 1: Customer Discovery & Booking Portal

- **Mobile-First Phone Auth:** Users authenticate directly through their 10-digit Indian mobile number (+91) and instant OTP verification without entering an email address.
- **Geospatial Proximity Discovery:** Users locate nearby marriage halls, lawns, and banquet venues filtered by city, pincode, and distance radius via Leaflet map integration.
- **3D Spatial Virtual Tour:** Users can pan, rotate, and navigate inside a 3D simulated banquet hall to inspect stage dimensions, lighting fixtures, and seating arrangements using Three.js.
- **Lightbox Photo Gallery:** High-resolution multi-image viewer with thumbnail navigation and full-screen preview.
- **Real-Time Calendar Availability:** Immediate visual status of dates: Green indicates available dates, while Red indicates dates already reserved or blocked.
- **Advance Token Payment:** Calculation of advance booking token (typically 10%–20%) with direct payment checkout and server-generated PDF confirmation.

Module 2: Venue Owner & Vendor Management Suite

- **Multi-Category Onboarding:** Venue owners register under one of 11 distinct event categories with domain-specific input fields.
- **Dynamic Profile & Pricing:** Owners update per-plate vegetarian/non-vegetarian rates, full-day lawn rental fees, seating capacities, and included amenities.
- **Multi-Photo Gallery Manager:** Powered by Multer, owners can upload and organize primary cover thumbnails and multiple interior/exterior gallery photos.
- **Offline Date Blocker Engine:** Venue owners can block specific calendar dates booked through offline channels in one click, preventing double-bookings.
- **Business Email Verification:** Owners submit corporate emails and GST/PAN numbers to obtain the trusted "Shield Verified Partner" badge.

Module 3: Superadmin Master Control Center

- **Executive KPI Dashboard:** Real-time metrics tracking Gross Booking Volume (GBV), total registered venues, active users, and platform revenue.
- **Vendor KYC Document Audit:** Administrators inspect uploaded business certificates and verify or suspend vendor accounts.
- **Commission Settlement Engine:** Configures platform service commission rates (ranging from 3% to 10%) on advance booking transactions.
- **Review Moderation:** Tools to audit customer ratings and filter out inappropriate or fraudulent reviews.
- **System Telemetry & Audit Logs:** Live monitoring of database connection state, API response latencies, and critical system events.

12. THE 11 SUPPORTED BUSINESS CATEGORIES DEEP-DIVE

#	CATEGORY NAME	EVENT SCOPE & APPLICATION	PRIMARY PRICING MODEL
1	Marriage Hall / Venue	Traditional weddings, mandap rituals, baraat receptions.	Full Day Rental + Plate Charges
2	Banquet Hall	Engagements, sangeet, corporate seminars, parties.	Time Slot / Per Plate
3	Marriage Garden / Lawn	Large outdoor weddings (1,000–5,000+ guests), tenting.	Lawn Rental per Event Day
4	Party Hall	Birthdays, anniversaries, kitty parties, reunions.	Hourly Package / Minimum Cover
5	Event Venue	Trade exhibitions, cultural conventions, public expos.	Daily Commercial Lease
6	Hotel & Resort	Destination weddings, residential stays, pool lawns.	Room Inventory + Venue Package
7	Restaurant & Dining	Family get-togethers, roka ceremonies, business dinners.	Per-Head Buffet / Set Menu
8	Catering Service	Live food counters (North/South Indian, Chinese).	Per Plate (Veg & Non-Veg Matrix)
9	Decoration & Planner	Stage floral styling, LED entrance arches, haldi themes.	Theme Package + Custom Add-ons
10	Photography / Film	Cinematic 4K films, pre-wedding shoots, drone captures.	Per Event Day / Full Package
11	Other Event Businesses	DJ sound systems, vintage bridal cars, dhol tasha.	Fixed Service Token Booking

13. DATABASE DESCRIPTION & MONGOOSE SCHEMAS

The platform utilizes MongoDB Atlas with Mon
backend/data/. Below are the primary collections

COLLECTION	FUNCTIONAL PURPOSE	IMPORTANT FIELDS
users	Customer accounts & authentication	id, name, mobile (Unique), role, isVerified, savedHalls
owners	Vendor commercial profiles & KYC	id, userId, businessName, businessCategory, email, gstNumber
halls	Venues, pricing, capacity, media	id, ownerId, name, startingPrice, capacityMax, coverImg, images, blockedDates
bookings	Reservations & payment status	id, bookingCode, customerId, serviceId, eventDate, advanceAmount, status
reviews	Customer star ratings & feedback	id, bookingId, serviceId, customerId, rating (1-5), comment, photos
transactions	Razorpay payment verification	id, bookingId, customerId, amount, razorpayPaymentId, razorpayOrderId
categories	11 Supported event categories	id, slug (Unique), name, icon, description, count

14. FRONTEND & BACKEND SYSTEM ARCHITECTURE

A. Frontend System Architecture

- **Rendering Engine:** Semantic HTML5 structure coupled with component-driven Vanilla CSS and modular ES6 JavaScript.
- **Virtual Walkthrough Subsystem:** Three.js renders a 3D procedural banquet hall utilizing standard PerspectiveCamera, WebGLRenderer, directional lighting, and touch/mouse orbit controls.
- **Session State:** Client sessions are maintained via JWT tokens stored in localStorage, accompanied by HTTP request interceptors for Authorization Bearer headers.
- **Responsive Grid:** CSS Grid and Flexbox with responsive breakpoints ensuring smooth rendering across screens from 320px smartphones to 4K displays.

B. Backend System Architecture

- **Routing Pipeline:** Modular Express router controllers mounted under /api/* endpoints with dedicated middleware chains.
- **Security Middleware:** Deep request payload sanitization (sanitizeInput), Cross-Origin Resource Sharing (cors), and rate limiters.
- **Media Ingestion:** Multer disk storage pipeline validates file MIME types (JPEG, PNG, WebP) and persists images to frontend/assets/uploads/.
- **Dynamic PDF Pipeline:** Server-side PDFKit stream engine programmatically renders multi-page vector invoices with exact bounding box constraints.
- **Failover Persistence:** Database connectors check live MongoDB connectivity; if disconnected, calls gracefully redirect to localized atomic JSON stores without crashing.

15. SYSTEM WORKFLOW & LIFECYCLE

The complete operational sequence follows 7 well-defined transitions:

- **Step 1 (Discover Venue):** Customer enters City/Category and filters listings by plate budget and capacity.
- **Step 2 (Inspect Venue):** Customer launches Three.js 3D Virtual Tour and browses high-res multi-image gallery.
- **Step 3 (Check Availability):** System queries database blockedDates array; displays Green (Available) or Red (Blocked).
- **Step 4 (Frictionless Auth):** Customer enters 10-digit mobile number (+91); receives instant 6-digit SMS OTP.
- **Step 5 (Advance Payment):** Razorpay UPI checkout initialized; server verifies HMAC-SHA256 signature.
- **Step 6 (Confirm Booking):** Booking record created; calendar date automatically locked to prevent schedule clashes.
- **Step 7 (Invoice & Alert):** PDFKit generates vector digital receipt; SMS alert dispatched to venue owner.

16. SECURITY HARDENING & DATA PRIVACY

- **Stateless JWT Auth:** API endpoints verify incoming Authorization: Bearer <token> headers signed using strong 64-character server secrets.
- **Bcrypt Cryptographic Hashing:** Vendor and administrative passwords undergo 10 rounds of salted Blowfish encryption via bcryptjs. No plain-text passwords ever exist in memory or database records.
- **NoSQL & XSS Sanitization:** Custom recursive sanitization middleware strips hazardous operators ($gt, $where, $regex) and escapes HTML entities from incoming payloads.
- **HMAC-SHA256 Webhook Verification:** Razorpay payment callbacks mathematically verify cryptographic signatures before confirming any booking.
- **Rate Limiting Shield:** Sensitive endpoints (OTP generation and login attempts) enforce strict IP-based rate limiting to prevent brute-force attacks and SMS quota depletion.
- **Enterprise Security Headers:** Implements standard security headers including X-Content-Type-Options: nosniff and X-Frame-Options: SAMEORIGIN.

17. ADVANTAGES OF THE PROPOSED SYSTEM

- **Tailored for India:** Pure mobile-phone authentication removes email friction and drives conversion rates.
- **Substantial Cost Savings:** Eliminates 10%–20% broker commission fees by fostering direct customer-to-vendor transactions.
- **Zero Scheduling Clashes:** The synchronized real-time availability calendar prevents double-bookings.
- **Informed Customer Decisions:** 3D virtual walkthroughs enable families to inspect hall setups without spending hours traveling to physical sites.
- **High Operational Speed:** Vanilla JS frontend delivers high performance, low data consumption, and fast load times on mobile devices.
- **Fault-Tolerant Persistence:** Dual-engine architecture guarantees zero downtime even if remote cloud database links experience latency.

18. SYSTEM LIMITATIONS

- **3D Model Procedural Nature:** The current 3D WebGL viewer renders standardized interactive architectural floor templates rather than photogrammetric real-world 3D scans of every individual hall.
- **SMS Gateway Dependency:** SMS OTP delivery requires an active paid SMS gateway subscription (e.g., Twilio/Fast2SMS) in production environments; in local development, it operates in simulated console mode.
- **Offline Balance Settlement:** While advance token reservation is handled digitally via Razorpay UPI, remaining event balances are settled directly between owner and customer on-site.

19. FUTURE SCOPE & TECHNOLOGICAL ENHANCEMENTS

- **Photogrammetric Drone Scans:** Transitioning from procedural 3D models to real drone-scanned 3D point-cloud environments of actual marriage gardens.
- **Cross-Platform Mobile Apps:** Packaging the mobile web architecture into native Android and iOS applications using Flutter or React Native.
- **AI Event Budget Planner:** Integrating Google Gemini AI to automatically generate itemized event budget estimates and vendor bundles based on guest count.
- **Full Escrow Milestone Payments:** Introducing a multi-stage digital escrow payment system where funds are released to vendors upon milestone completions (e.g., Stage Setup Complete, Dinner Commenced).

20. CONCLUSION

Hall Khoj Finder provides a practical, full-stack engineering solution to address the longstanding inefficiencies of the Indian event venue booking market. By thoughtfully combining a mobile-first customer journey (phone-only authentication) with advanced technical capabilities like Three.js 3D WebGL spatial exploration, real-time calendar synchronization, automated PDF generation, and a fault-tolerant dual-database architecture, the project establishes a robust, highly scalable, and user-centric platform. The implementation strictly adheres to modern web engineering principles and is fully prepared for academic project evaluation and real-world commercial deployment.

21. PROJECT SUMMARY (VIVA QUICK-REFERENCE SHEET)

PROJECT PARAMETER	ACADEMIC VIVA SUMMARY SPECIFICATION
Project Name	Hall Khoj Finder
Domain	Web Engineering, E-Commerce Marketplace, 3D WebGL Computer Graphics
Architecture Pattern	Client-Server 3-Tier RESTful Architecture with Mobile-First Responsive PWA
Frontend Technologies	HTML5, Vanilla CSS3, Modern ES6+ JavaScript, Three.js (3D), Leaflet.js (Maps)
Backend Framework	Node.js Runtime (v24.x) with Express.js (v5.x REST API Engine)
Database Layer	MongoDB Cloud Atlas + Mongoose ODM (with Local Atomic JSON Failover)
Security & Cryptography	JWT (HS256), Bcrypt (10 Salt Rounds), HMAC-SHA256 Webhook Verification
Key Innovations	1. Phone-only OTP login 2. Three.js 3D Virtual Tour 3. 1-Click Date Blocker 4. Dynamic PDFKit Invoicing 5. 11 Supported Categories
Target User Roles	Event Seekers (Customers), Venue Owners (Vendors), System Administrators